

Elaina Lutz

(484) 681-0820 | elaina.lutz@temple.edu | www.linkedin.com/in/elaina-lutz | U.S. Citizen

EDUCATION

Temple University – College of Engineering

Philadelphia, PA

Bachelor of Science

Expected Graduation: May 2028

Major: Electrical and Computer Engineering, Electrical Concentration | **GPA:** 3.99/4.00

Honors: Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026 / Provost's Tuition Scholarship / Honors Program

Relevant Coursework: Power Electronics, Digital Circuit Design, Signals and Systems, Computation II

WORK EXPERIENCE

Neural Engineering Data Consortium

Philadelphia, PA

Software Engineer

August 2026 – Present

- Develop machine learning tools for biofilm image analysis and architect a web-based diagnostic application.
- Build automated training pipelines, biostatistics programs, and turn-key scripts for model training.

Temple University College of Engineering

Philadelphia, PA

Engineering Computation I Peer Mentor

December 2025 – Present

- Evaluate and grade student submissions for code correctness, algorithmic efficiency, and analytical rigor, providing structured feedback on testable code practices and computational problem-solving.

Introduction to Engineering and Engineering Technology Teaching Assistant

August 2025 – December 2025

- Assisted first-year engineering students with technical problem solving and assignment verification, conducting weekly office hours providing guidance on systems-based design thinking and engineering fundamentals.

Constellation Energy – Crane Clean Energy Center

Middletown, PA

Programs & CMO Electrical Engineering Intern

June 2026 – August 2026

- Led verification workflow for breaker refurbishment and replacement operations, coordinating across site departments to ensure compliance with IEEE/NERC standards and documenting findings in technical reports.
- Assisted with aging management programs to ensure regulatory compliance and equipment reliability.

Strategic Electrical Engineering Intern

June 2025 – August 2025

- Performed inverter ID verification and asset management, validating equipment prior to operational restart.
- Conducted plant walkdowns to assess equipment readiness and analyzed electrical loading data for safe plant operations.

PROJECTS

Arcade Game Engine

April 2026

- Built a distributed game platform with Python and C++ servers communicating via TCP/JSON, featuring custom data structures optimized for performance and memory efficiency.
- Developed Pygame client with real time chat and game browsing UI, engineered polymorphic serialization for flexible wire formats, debugged complex network communication, and player state synchronization across remote connections.

Musical Light Show Circuit

April 2026

- Designed a 4-stage analog signal processing system with op-amp mixing, adjustable gain control, and active frequency separation using RC low-pass and high-pass filters derived from FFT spectral analysis.
- Engineered peak detection and BJT-based LED driver circuits with current limiting, validating performance through NI Multisim simulation and oscilloscope-based hardware testing.

LEADERSHIP & ACTIVITIES

Eta Kappa Nu (HKN) – IEEE Honor Society

April 2026 - Present

Power and Energy Society (PES), IEEE Branch

Philadelphia, PA

Marketing Director

March 2026 – Present

- Organized an industry panel connecting Temple IEEE PES members with Constellation Energy engineers.

Women in Engineering (WIE), IEEE Branch

Philadelphia, PA

Marketing Chair

Sept 2025 – Present

- Develop marketing materials and organize professional resources for chapter members.

Theta Tau – Professional Engineering Fraternity

Philadelphia, PA

Treasurer

January 2026 – Present

Institute of Electrical and Electronics Engineers

Philadelphia, PA

Member

Sept 2024 – Present

SKILLS

Software: Python, C/C++, MATLAB, Linux/Unix Command Line, NumPy, NI Multisim, KiCad, AutoCAD, GitHub, Bash, Inventor 2025, Matplotlib, AWS, VS Code, EMACS, draw.io, Scopy, Slurm, Microsoft Office Suite

Hardware: Circuit Design & Breadboard Prototyping, Embedded Systems (Arduino, Micro:bit), Test & Measurement (Oscilloscopes, Multimeters), Soldering & PCB Assembly